

Con. 6595-13.

GS-6621

(3 Hours)

[Total Marks : 100]

N.B. : (1) Question No. 1 is **compulsory**.(2) Attempt any **four** out of remaining **six** questions.(3) Assume **suitable** data wherever **necessary**.(4) Answer to **each** new question to be started on **fresh** page.

1. (a) Explain different Instruction formats with suitable example. **10**
(b) Define following terms :— **10**
 - (i) Computer organization.
 - (ii) Computer Architecture.
 - (iii) MDR.
 - (iv) PC.
 - (v) SP.
2. (a) Explain Instruction cycle with interrupt execution in detail. **10**
(b) Compare and explain static and dynamic data flow computers. **10**
3. (a) Explain IEEE-754 formats. **10**
(b) Explain cache memory mapping techniques with example. **10**
4. (a) Explain Micro-programmed control unit in detail. **10**
(b) Explain 6-stage Instruction execution with pipelined processor. **10**
5. (a) Explain types of memories based on the hierarchy of speed and size. **10**
(b) Define “(Input/Output) I/O Module”. State the difference between Programmable and Non-programmable devices with suitable examples. **10**
6. (a) Compare RISC and CISC processors. **10**
(b) Explain Interleaved Memory with low-order and high-order Interleaving. **10**
7. Write short notes on any **four** of the following :— **20**
 - (a) RAID Memory.
 - (b) Booth’s Algorithm.
 - (c) MIMD and SIMD.
 - (d) Paging and Segmentation.
 - (e) Page Replacement Algorithm.